DECENT LOCALLY FINITE TYPE SPACES OVER UNIVERSALLY CATENARY BASES ARE CATENARY

THE STACKS PROJECT AUTHORS

Editorial note. Counted span is the author's own proof environment, source lines 5823-5834, immediately following the statement at 5815-5821.

Editorial note. Three cross-chapter pointers (Topology x3) and two same-chapter line-indexed stubs are recorded in assembly.external_references.

Allowed standard prerequisites (editorial).

- Catenarity is local on the base. Topology, Lemma topology-lemma-catenary, cited at proof lines 5823-5824. Reduces to the case where S carries a dimension function.
- Dimension functions exist locally. Topology, Lemma topology-lemma-locally-dimension-function, cited at proof lines 5825-5826. After localization, S has a dimension function.
- Dimension functions transfer to decent locally finite type spaces. Lemma lemma-scheme-with-dimension-function of this chapter, cited at proof lines 5827-5828. |X| inherits a dimension function from S.
- Sober spaces with dimension functions are catenary. Topology, Lemma topology-lemma-dimension-function-catenary, cited at proof lines 5831-5833. The terminal criterion: |X| sober plus dimension function gives catenary.
- Reasonable spaces have sober underlying topologies. Proposition proposition-reasonable-sober of this chapter, cited at proof line 5830. |X| is sober – the last hypothesis the terminal criterion needs.

Source reference. Original lemma 0ED5: sources/stacks-decent-spaces/pinned-original-source.tex, source lines 5783–5783. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original proposition 03K6: sources/stacks-decent-spaces/pinned-original-source.tex, source lines 2915–2915. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 02I2: focused-seeds/geometry/S9-decent-universally-catenary-scheme/ref-tags.txt, source lines 5–5. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Date: stacks-project:decent-spaces.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Source reference. Original reference 02IA: focused-seeds/geometry/S9-decent-universally-catenary-scheme/ref-tags.txt, source lines 6–6. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 02IC: focused-seeds/geometry/S9-decent-universally-catenary-scheme/ref-tags.txt, source lines 7–7. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Lemma 0.1. *Let S be a locally Noetherian and universally catenary scheme. Let X be an algebraic space over S such that X is decent and such that the structure morphism $X \to S$ is locally of finite type. Then X is catenary.*

Proof. The question is local on S (use Topology, Lemma 02I2). Thus we may assume that S has a dimension function, see Topology, Lemma 02IC. Then we conclude that $|X|$ has a dimension function by Lemma 0ED5. Since $|X|$ is sober (Proposition 03K6) we conclude that $|X|$ is catenary by Topology, Lemma 02IA. □